stable polymorphism. As far as the maths are concerned, an exactly equivalent ESS can be achieved without polymorphism as follows. If *every individual* is capable of behaving either like a hawk or like a dove in each particular contest, an ESS can be achieved in which all individuals have the same *probability* of behaving like a hawk, namely $\frac{7}{12}$ in our particular example. In practice this would mean that each individual enters each contest having made a random decision whether to behave on this occasion like a hawk or like a dove; random, but with a 7:5 bias in favour of hawk. It is very important that the decisions, although biased towards hawk, should be random in the sense that a rival has no way of guessing how his opponent is going to behave in any particular contest. It is no good, for instance, playing hawk seven fights in a row, then dove five fights in a row and so on. If any individual adopted such a simple sequence, his rivals would quickly catch on and take advantage. The way to take advantage of a simple sequence strategist is to play hawk against him only when you know he is going to play dove.

The hawk and dove story is, of course, naïvely simple. It is a 'model', something that does not really happen in nature, but which helps us to understand things that do happen in nature. Models can be very simple, like this one, and still be useful for understanding a point, or getting an idea. Simple models can be elaborated and gradually made more complex. If all goes well, as they get more complex they come to resemble the real world more. One way in which we can begin to develop the hawk and dove model is to introduce some more strategies. Hawk and dove are not the only possibilities. A more complex strategy which Maynard Smith and Price introduced is called *Retaliator*.

A retaliator plays like a dove at the beginning of every fight. That is, he does not mount an all-out savage attack like a hawk, but has a conventional threatening match. If his opponent attacks him, however, he retaliates. In other words, a retaliator behaves like a hawk when he is attacked by a hawk, and like a dove when he meets a dove. When he meets another retaliator he plays like a dove. A retaliator is a *conditional strategist*. His behaviour depends on the behaviour of his opponent.

Another conditional strategist is called *Bully*. A bully goes around behaving like a hawk until somebody hits back. Then he immediately runs away. Yet another conditional strategist is *Prober–retaliator*. A